Aedin Lai

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Education

Northeastern University Expected May 2029

Candidate for Bachelor of Science in Electrical & Computer Engineering — Boston, MA
Relevant coursework: Embedded Design, Computing Fundamentals for Engineers, Physics **2**, Calculus **2**

Experience

SingleStore June – September 2026

Data Center Infrastructure Intern — Boston, MA
- Led hardware deployment for a phased data center expansion, scaling a private cloud platform from **160** to **300+** VMs
- Replaced a single-push rollout with a phased "two racks at a time" plan, bringing three cohorts of **40–50** VMs online while limiting disruption to live workloads
- Diagnosed a Linux kernel storage defect and a multi-week NFS instability, restoring full fleet capacity
- Built a Go audit tool and REST API that checks power supply, memory, NIC, and drive health across the server fleet via iDRAC
- Supported a full Oakland-to-Boston data center relocation, migrating the platform from bare-metal to virtualized infrastructure

PEACE Foundation, Yale University School of Medicine 2023 – 2024

Research Intern — New Haven, CT
Supervisor: Dr. Kun Liu, Associate Professor of Sports Neuroscience
- Designed, built, and improved a full-body balance measurement device to study cognition and movement in neurological patients
- Automated data collection using custom hardware across ASD, Alzheimer's, Parkinson's, and schizophrenia patient populations
- Presented at an international conference (ISEC **2024**); co-authored a study accepted to JAMA **(2025)**

Projects

aedinlai.com — personal site & project hub February 2025 – Present

JavaScript, Cloudflare Workers, SQL, GitHub Actions
- Built a framework-free single-page site hosting my live projects and a daily auto-generated news feed
- Engineered a gym occupancy tracker that records live counts every **10** minutes into a serverless SQL database
- Designed a serverless SQL API for a shared pizza-pricing app, with per-IP rate limiting and owner-only edits

Sandbit — pixel physics engine September 2026

Rust, WebAssembly, WebGL
- Built a **5,700**-line Rust/WebAssembly falling-sand engine simulating **27** materials, including electrical conduction
- Implemented rigid-body physics with contact manifolds, buoyancy, and rope and motor joints for pixel objects
- Verified with **30** physics tests, fuzzing, and a headless player that runs all **15** levels across **5** random seeds

One-Inch Punch Arcade Machine September – December 2025

Cornerstone of Engineering — laser sensors, electronics, mechanical design
- Designed a dual-laser-sensor trigger and wrote scoring logic that converts swing velocity into a live score
- Led the full build, from electronics to a pendulum-and-magnet auto-reset, and validated it with **80+** full-force stress tests
- Identified a paddle material limit during a full day of expo play and proposed a rubber-composite redesign

Hallway Marathon — event website November 2025

HTML, CSS, JavaScript
- Built a framework-free event site with live betting odds and competitor stats, run live for a multi-university race

More projects at aedinlai.com and github.com/Aedin-ctrl

Leadership & athletics

Northeastern University Club Fencing 2025 – Present

Captain
- Recruit new fencers and design practices; anchor the epee squad at the **2026** Collegiate Club National Championship

Marx Fencing Academy 2018 – 2025

Competitive epee fencer
- Nationally ranked **#103** in the U.S.; Top-**16** World Cup finish; All-American **2021–2025**

Skills

Programming	Python, Java, C#, Rust, SQL, MATLAB, JavaScript, HTML/CSS
Hardware & CAD	Quartus Prime, Raspberry Pi, sensor integration, SolidWorks, Fusion 360, AutoCAD, laser cutting
Tools	Git, GitHub Actions, Linux, Cloudflare Workers, WebAssembly, VS Code

Live projects: aedinlai.com/gatecraft, /sandbit, /hand-playground, /doodle-life, /games, /gym, /broadcast, and football.aedinlai.com